CS 5180 – Assignment #4

Maximum Points: 100 pts.

Bronco ID: |__|__|__|__|__|__|__|__|__|

Last Name: _____

First Name: _____

Note 1: Your submission header must have the format as shown in the above-enclosed rounded rectangle.
Note 2: Homework is to be **done individually**. You may discuss the homework problems with your fellow students, but you are NOT allowed to copy – either in part or in whole – anyone else's answers.
Note 3: Your deliverable should be a **.pdf file** submitted through **Gradescope** by the deadline. Do not forget to **assign a page to each of your answers** when making a submission. In addition, source code (.py files) should be added to an **online repository (e.g., GitHub)** to be downloaded and executed later.
Note 4: All submitted materials must be legible. Figures/diagrams must be of good quality.
Note 5: Please use and check the Canvas discussion for further instructions, questions, answers, and hints. The bold words/sentences provide information for a complete or accurate answer.

1. [20 points]. Given the documents and query below, answer the following questions.

Query	Vector
q	(1,0,0,1,1)

Document	Vector	Relevance
d_1	(1,1,0,0,1)	R
d_2	(0,1,1,1,0)	NR
d_3	(1,0,1,0,1)	R
d_4	(0,0,1,1,1)	NR
d_5	(1,0,0,1,0)	NR
d_6	(1,1,1,0,0)	R
d_7	(0,1,0,0,1)	NR
d_8	(1,0,1,1,1)	R

   a. [10 points]. Compute the BIM term weight for each query term.
   b. [5 points]. Compute the BIM score for each document.
   c. [5 points]. Rank the documents from highest score to lowest score.

2. [20 points]. Given the data provided, answer the following questions. Use $k_1 = 1.2$, $b = 0.75$, $k_2 = 0$.

Query	Vector
q	(1,0,1)

Document	Vector
d_1	(3,2,0)
d_2	(0,1,4)
d_3	(0,3,0)
d_4	(0,2,1)
d_5	(1,0,0)

a. [10 points]. Compute the BM25 IDF-like weight for each query term.
b. [5 points]. Compute the BM25 score for each document.
c. [5 points]. Rank the documents from highest score to lowest score.

3. [20 points]. After expanding the dataset of question 3 to include labels, answer the following questions. Use $k_1 = 1.2$, $b = 0.75$, $k_2 = 0$.

Query	Vector
q	(1,0,1)

Document	Vector	Relevance
d_1	(3,2,0)	R
d_2	(0,1,4)	NR
d_3	(0,3,0)	NR
d_4	(0,2,1)	R
d_5	(1,0,0)	R

a. [10 points]. Compute the BM25 relevance-weighted (IDF-like) weight for each query term.
b. [5 points]. Compute the BM25 score for each document.
c. [5 points]. Rank the documents from highest score to lowest score.

4. [20 points]. Language Model (TBD).

5. [20 points]. You are provided with **100 documents, 5 queries, and binary relevance judgments** for each query. Using this data, complete the Python program (bm25_search.py) to build a BM25-based search engine and evaluate its performance.

Requirements:

- Preprocess the documents and queries by removing stopwords only.
- Use the Python library *rank_bm25* to build a BM25 model from the document collection.
- For each query, compute the BM25 score of every document and rank the documents in descending order of score.
- Using the relevance judgments, compute Average Precision (AP) for each query.
- Sort the queries in descending order of their AP values and report the ranking.

Important Note: Answers to all questions should be written clearly, concisely, and unmistakably delineated. You may resubmit multiple times until the deadline (the last submission will be considered).

NO LATE ASSIGNMENTS WILL BE ACCEPTED. ALWAYS SUBMIT WHATEVER YOU HAVE COMPLETED FOR PARTIAL CREDIT BEFORE THE DEADLINE!